

Meal Planning and Diet Management System

Architecture Selection & Justification

Architecture Selection: We chose **Microservices Architecture** for the Meal Planning and Diet Management System.

1. Architectural Choice

The system is divided into independent services such as User Profile & Preferences, Dietary Preference, Meal Subscription, Payment & Billing, Meal Planning, and Meal Recommendation. Each service handles a specific business function and communicates with other services through defined interfaces.

2. Reasons for the Choice

Reason 1 – Independent functionality: The system contains several separate features such as meal planning, recommendations, subscriptions, user preferences, and payments. Microservices allow these functions to be developed and maintained independently.

Reason 2 – Scalability and flexibility: Frequently used services, such as Meal Recommendation or Meal Planning, can be scaled independently without scaling the entire application. Services can also be updated without affecting unrelated modules.

3. Security Advantage

Microservices provide service-level separation. Sensitive operations such as Payment & Billing can be isolated from meal and profile services, with controlled API access between services. This limits unnecessary access to sensitive payment information.

4. Performance Benefit

Independent services can be scaled according to demand. For example, the Meal Recommendation Service can handle a large number of recommendation requests without requiring the User Profile or Payment services to be scaled at the same time.

Conclusion: Microservices Architecture is suitable because it provides modularity, independent scalability, controlled service communication, and separation of sensitive operations for the Meal Planning and Diet Management System.